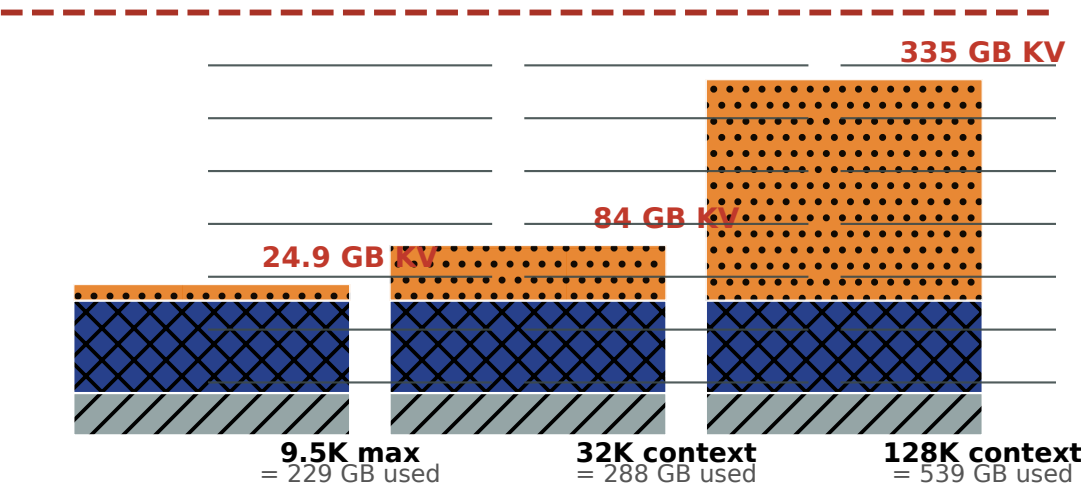


Memory Tetris: how the 8×H100 host (640 GB) fills with context

640 GB = 8 × 80 GB ranks (horizontal rules = 80 GB rank boundaries)



baseline (weights + runtime) is context-independent;
the KV cache is the lever that grows with context.
FP16 ~2.5 MB/token [ILLUSTRATIVE][DERIVED].

AGGREGATE RESIDENCY SCREEN ≠ PER-RANK FIT GUARANTEE
(total < 640 GB is a first-order check;
sharding / fragmentation still validated per GPU)

- KV cache (grows w/ context)
- weights 140 GB
- runtime / NCCL ~64 GB